SOFTWARE REQUIREMENTS SPECIFICATION

TaskFlow Pro

AI-Powered Universal Team Task Management Platform

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Course Assignment 2 - Project 1

Task 1: Team Selection

Team Members:

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Task 2: Software Requirements Specification

1. Preface

This Software Requirements Specification (SRS) document describes the requirements for TaskFlow Pro — an AI-powered, universal team task management platform. The

system is designed to serve teams across any domain or industry, including software development, marketing, human resources, design, education, finance, and operations. While the platform supports specialized features such as an integrated code compiler for software teams, its core mission is to be a general-purpose workspace where team leaders can assign any type of task — from writing a market research report or designing a graph, to solving a coding problem or producing a presentation — all within a single, unified environment.

This document is intended for developers, project managers, stakeholders, and evaluators involved in the development and assessment of the system.

Version History:

- Version (March 2026): Initial release of the SRS document covering all core modules including authentication, team management, universal task assignment, optional integrated code compiler, AI-assisted features, and reference material sharing.

This document follows the SRS structure as outlined in Figure 4.17 of Software Engineering by Ian Sommerville, 10th edition.

2. Introduction

2a. Purpose

The purpose of TaskFlow Pro is to provide a centralized, AI-enhanced workspace where team leaders (managers) can assign, track, and manage any kind of task for their team members — regardless of the department or nature of the work. Unlike tools that are narrow in focus (e.g., platforms only for coding challenges, or tools only for project ticketing), TaskFlow Pro is built as a universal task assigner inspired by the flexibility of Google Classroom. A manager can assign a software engineer to fix a bug, ask a designer to create a graph or infographic, instruct an HR associate to draft a policy document, request a marketing analyst to prepare a campaign report, or set a finance intern to complete a spreadsheet analysis — all from the same interface.

The platform also integrates a built-in code compiler as a specialized feature for software and engineering teams, so coding tasks can be executed and submitted directly within the platform. An AI layer, powered by machine learning models (via open-source models or external APIs), assists managers and members through intelligent task recommendations, automated feedback on deliverables, and deadline-aware notifications. The goal is to eliminate tool fragmentation and give every kind of team a single, powerful place to collaborate and deliver.

2b. Scope

TaskFlow Pro is a web-based application that supports the following high-level functionalities:

- User authentication: registration, login, and role management (Manager / Team Member).
- Team management: creating a new team or joining an existing team via a unique invite code.
- Universal task assignment: managers can assign any type of task to team members, including (but not limited to): coding problems, document writing, graph or chart creation, presentation design, marketing briefs, HR reports, research tasks, and data analysis. Each task includes a title, description, deadline, optional task type tag, and optional reference materials.
- Flexible task type tags: tasks can be tagged by type such as Coding, Design, Document, Research, Marketing, HR, Finance, or Custom — allowing the manager to categorize work across any department.
- Integrated online compiler (optional, for coding tasks): members can write, run, and submit code in Python, C++, Java, JavaScript, and other languages directly in the browser when assigned a coding-type task.
- AI-assisted features: automated feedback on submitted deliverables powered by machine learning, intelligent task suggestions, and smart deadline reminders.
- Manager dashboard: full visibility of all team members, assigned tasks, submission statuses, task type distribution, and upcoming deadlines.
- Reference material sharing: managers can upload files, paste URLs, or link resources (PDFs, videos, articles, design briefs) to any task or share them with the entire team — similar to Google Classroom.
- Submission and review workflow: team members submit their completed work (code, documents, files, or links) and the manager can review and provide feedback, with AI assistance.

The system is accessible via modern web browsers and does not require any local installation by end users. It is designed to be domain-agnostic, meaning any team — whether in a corporate, educational, or freelance context — can use it effectively.

3. Glossary

- Manager: A user with elevated privileges within a team who can create and assign tasks of any type, set deadlines, share reference materials, and monitor team progress. Also referred to as Team Leader.
- Team Member: A standard user who joins a team via an invite code, receives assigned tasks, submits completed work, and views feedback.
- Invite Code: A unique alphanumeric string generated when a new team is created, used by other users to join that specific team.

- Task: A unit of work assigned by a manager. Tasks are not limited to coding — they can include writing documents, creating graphs, preparing reports, conducting research, designing materials, and any other type of deliverable.
- Task Type Tag: A label applied to a task to categorize its nature, e.g., Coding, Design, Document, Research, Marketing, HR, Finance, or Custom. Used for filtering and reporting on the dashboard.
- Compiler: An integrated online code execution environment available for coding-type tasks, supporting multiple programming languages including Python, C++, Java, and JavaScript.
- Submission: A team member's completed response to a task, which may take the form of submitted code, an uploaded file, a typed document, a shared link, or any other deliverable format.
- ML Model: A machine learning model integrated via API (e.g., OpenAI GPT-4, HuggingFace) used to provide automated feedback on submissions, suggest tasks, and flag deadline risks.
- Deadline: The date and time by which a task must be completed and submitted by the assigned member(s).
- Reference Material: Documents, links, PDFs, images, videos, or any resource attached by the manager to a task or shared within the team workspace to assist members in completing their work.
- Dashboard: A visual summary interface available to the manager showing team-wide task progress, statuses, task type breakdown, and deadline timelines.
- Workspace: The team's shared area within TaskFlow Pro, encompassing all tasks, members, materials, and activity for that team.

4. User Requirements Definition

4a. Functional Requirements

The following functional requirements describe what the system must do from the user's perspective:

1. Users shall be able to register an account by providing a username, email address, and password.
2. Users shall be able to log in using their email and password, and log out at any time.
3. A logged-in user shall be able to create a new team, becoming its Manager, or join an existing team by entering a valid invite code.
4. A Manager shall be able to create and assign any type of task — including but not limited to coding problems, document writing, graph creation, marketing reports, HR tasks, research briefs, and design work — to one or more team members, with a title, description, task type tag, deadline, and optional reference materials.

5. A Team Member shall be able to view all their assigned tasks, including task descriptions, type tags, deadlines, and any attached reference materials.
6. For coding-type tasks, a Team Member shall be able to write, execute, and submit code using the built-in online compiler for supported languages.
7. For non-coding tasks, a Team Member shall be able to submit their work by uploading a file, entering a text response, or providing a link to an external deliverable.
8. A Manager shall be able to view all team members and the status of every assigned task from a centralized dashboard, filterable by task type, member, or deadline.
9. A Manager shall be able to share reference materials — including documents, URLs, videos, and design briefs — with the entire team or attached to a specific task, similar to Google Classroom.
10. The system shall use an AI model to generate automated feedback on task submissions, applicable to any task type (e.g., code review for coding tasks, writing quality analysis for document tasks, and relevance scoring for research tasks).
11. The system shall notify team members of upcoming deadlines via in-app alerts.

4b. Non-Functional Requirements

12. Usability: The interface shall be intuitive enough that a new user from any background — technical or non-technical — can register, join a team, and submit their first task within 10 minutes without external guidance.
13. Performance: Task submission and AI feedback generation shall complete within 15 seconds under normal system load.
14. Availability: The system shall be available 99% of the time on a monthly basis.
15. Security: All passwords shall be stored using bcrypt hashing. All communication shall be encrypted using HTTPS/TLS.
16. Scalability: The system shall support at least 500 concurrent users without performance degradation.

5. System Requirements Definition

5a. Functional Requirements

The following system-level functional requirements specify detailed technical behaviors:

17. The system shall authenticate users via JWT (JSON Web Token) tokens with a validity period of 24 hours, refreshable on active session.
18. The system shall generate a unique 8-character alphanumeric invite code upon team creation and store it in the database linked to that team record.

19. When a user submits a valid invite code, the system shall add them to the corresponding team with the Team Member role and grant access to the team workspace.
20. The system shall store tasks with the following attributes: task_id, title, description, task_type_tag (enum: Coding, Design, Document, Research, Marketing, HR, Finance, Custom), language (nullable, required only for Coding type), deadline, assigned_to (list of user IDs), created_by (manager ID), status (Pending / In Progress / Submitted / Reviewed / Overdue), and attached_materials.
21. The task submission module shall accept the following input formats based on task type: source code files or text (Coding), uploaded file attachments up to 50 MB (Document, Design, HR, Finance, Marketing), plain text responses (Research, Custom), and external URLs (any type).
22. For Coding-type tasks, the system shall send submitted code to a sandboxed execution environment (e.g., Judge0 API or Docker-based runner) and return stdout, stderr, and execution time to the user.
23. The ML integration module shall send task submissions to an AI API (e.g., OpenAI GPT-4 or a domain-appropriate HuggingFace model) and receive structured feedback including: a quality/relevance score (0-100), key issues identified, and actionable improvement suggestions. The feedback prompt shall be adapted based on the task type tag.
24. The manager dashboard module shall query the database for all tasks in the manager's team and return: total task count, count by status, count by task type tag, overdue count, and a per-member completion rate.
25. The reference materials module shall support file uploads up to 50 MB per file and URL linking, storing materials in association with either a specific task or the team workspace, and displaying them in a Materials Panel accessible to all team members.
26. The system shall automatically transition a task's status to Overdue when the current server timestamp exceeds the task deadline and the status remains Pending or In Progress.
27. The notification module shall trigger an in-app reminder alert for assigned members 24 hours before a task's deadline.
28. The system shall enforce Role-Based Access Control (RBAC): Managers may create tasks, assign members, upload materials to the workspace, and view all submissions; Team Members may only view their own tasks, submit work, and access shared materials.
29. The system shall maintain an immutable submission audit log storing: submission timestamp, submitting user ID, submitted content reference, and AI feedback result for every task submission.

5b. Non-Functional Requirements

30. Performance: The backend API shall respond to at least 95% of requests within 500ms under a load of up to 200 concurrent requests per second.
31. Scalability: The system shall use a modular, horizontally scalable architecture, allowing the compiler service, the ML feedback service, and the core task service to scale independently.
32. Security: All code submitted for execution shall be run in an isolated, resource-limited sandbox with no network access to prevent arbitrary system exploitation or denial-of-service attacks.
33. Reliability: The system shall maintain automated database backups every 24 hours and support point-in-time recovery to within 1 hour of any failure.
34. Maintainability: The codebase shall maintain at least 70% unit test coverage for all backend service modules.
35. Compatibility: The frontend shall be fully functional on the latest two major versions of Chrome, Firefox, Safari, and Edge on both desktop and mobile screen sizes.
36. Extensibility: The task type tag system shall be designed as an open enumeration, allowing new task categories (e.g., Legal, Operations, Sales) to be added without requiring changes to the core task schema or UI logic.

6. System Models

6a. Use Case Models

Use Case 1: User Registration

Use Case ID: UC-01

Use Case Name: User Registration

Actors: Unregistered User (Primary)

Preconditions: The user has not previously registered. The registration page is accessible.

Postconditions: A new user account is created and stored in the system. The user is redirected to the login page.

Main Flow:

37. The user navigates to the registration page.
38. The user enters a username, valid email address, password, and confirms the password.
39. The user clicks the Register button.
40. The system validates that all fields are filled, the email is in valid format, and the password meets minimum criteria (at least 8 characters, one uppercase letter, one number).

41. The system checks that the email is not already associated with an existing account.
42. The system hashes the password using bcrypt and stores the new user record in the database.
43. The system displays a confirmation message and redirects the user to the login page.

Alternative Flows:

- 4a. If any field is empty or the email format is invalid, the system displays an inline error and highlights the offending field.
- 5a. If the email is already registered, the system displays: 'An account with this email already exists.'

Exceptions:

- Database connectivity failure: the system displays a generic error and logs the issue server-side.

Use Case 2: Create or Join a Team

Use Case ID: UC-02

Use Case Name: Create or Join a Team

Actors: Authenticated User (Primary)

Preconditions: The user is logged in and does not yet belong to a team.

Postconditions: The user is a member of a team, either as Manager (if creating) or as Team Member (if joining).

Main Flow - Create Team:

44. The user selects Create New Team.
45. The user enters a team name, optional team description, and selects the team's primary domain (e.g., Software, Marketing, HR, Education, Mixed/Other).
46. The system generates a unique 8-character alphanumeric invite code.
47. The system creates the team record and assigns the user the Manager role.
48. The system displays the invite code prominently so the manager can share it.

Main Flow - Join Team:

49. The user selects Join Existing Team.
50. The user enters the invite code provided by the manager.
51. The system validates the invite code against all active teams in the database.
52. If valid, the system adds the user to the team with the Team Member role.
53. The user is redirected to the team workspace.

Alternative Flows:

- 3a (Join). If the invite code is invalid or does not match any team, the system displays: 'Invalid invite code. Please check and try again.'

Use Case 3: Assign a Task to Team Members

Use Case ID: UC-03

Use Case Name: Assign a Task to Team Members

Actors: Manager (Primary)

Preconditions: The manager is logged in, belongs to a team, and at least one team member exists in the team.

Postconditions: A new task is created, stored, and associated with the selected team members. In-app notifications are sent to assigned members.

Main Flow:

54. The manager navigates to the Create Task page.
55. The manager fills in the task title and a detailed description.
56. The manager selects a task type tag from the available options: Coding, Design, Document, Research, Marketing, HR, Finance, or Custom.
57. If the task type is Coding, the manager additionally selects the target programming language and optionally enters test cases.
58. For all other task types (Design, Document, Marketing, HR, etc.), the manager provides any relevant instructions or expected output format in the description field.
59. The manager sets a deadline (date and time).
60. The manager selects one or more team members to assign the task to.
61. The manager optionally attaches reference materials: files (PDFs, images, spreadsheets, design briefs) or URLs (articles, videos, Google Docs links).
62. The manager clicks Assign Task.
63. The system validates all required fields and confirms the deadline is in the future.
64. The system saves the task record and sends in-app notifications to all assigned members.

Alternative Flows:

- 10a. If the deadline is in the past, the system displays: 'Deadline must be a future date and time.'
- 10b. If no team members are selected, the system displays: 'Please select at least one team member.'

Use Case 4: Submit Task Deliverable

Use Case ID: UC-04

Use Case Name: Submit Task Deliverable

Actors: Team Member (Primary), ML Feedback API (Secondary), Code Execution Service (Secondary, for Coding tasks only)

Preconditions: The team member is logged in, has at least one task assigned, and the task deadline has not passed.

Postconditions: The deliverable is stored, AI feedback is generated and displayed, and the task status is updated to Submitted.

Main Flow:

65. The team member opens an assigned task from their task list.
66. The team member reviews the task description, type tag, deadline, and any attached reference materials.
67. Depending on the task type, the team member prepares their deliverable: writes code in the embedded editor (Coding), types a written response (Document/Research), uploads a file (Design/HR/Marketing/Finance), or pastes an external link (any type).
68. For Coding tasks only: the team member may click Run to test their code; the system sends it to the sandboxed execution service and displays the output (stdout, stderr, execution time).
69. The team member clicks Submit.
70. The system saves the submission and sends the deliverable content (or a summary/excerpt for non-text files) to the ML feedback API, along with the task type tag as context for the AI model.
71. The ML API returns: a quality/relevance score, key issues identified, and improvement suggestions tailored to the task type.
72. The system displays the AI feedback to the member and updates the task status to Submitted.

Alternative Flows:

- 4a. If the code execution service is unavailable, the system displays a warning and allows the member to submit without running.
- 7a. If the ML API times out, the system stores the submission and marks the AI feedback as Pending, retrying in the background.

Exceptions:

- If the task deadline has already passed, the Submit button is disabled and the system displays: 'Deadline has passed. Submissions are no longer accepted for this task.'

Use Case 5: View Manager Dashboard

Use Case ID: UC-05

Use Case Name: View Manager Dashboard

Actors: Manager (Primary)

Preconditions: The manager is logged in and owns at least one team with at least one task created.

Postconditions: The manager has a current, comprehensive view of team activity and can take further actions.

Main Flow:

73. The manager navigates to the Dashboard.
74. The system queries all tasks and members associated with the manager's team.
75. The system displays aggregate statistics: total tasks, pending count, submitted count, reviewed count, overdue count, and a task type breakdown chart.
76. The system displays a task list table with columns: Task Title, Task Type, Assigned To, Deadline, Status.
77. The manager can filter the task list by status (All / Pending / Submitted / Overdue), by task type tag, or by specific team member.
78. The manager can click on any task to open its detail view, including all member submissions, AI feedback results, and attached materials.
79. The manager can click on a team member's name to view that member's individual progress page showing all their assigned tasks and completion rates.

Alternative Flows:

- 2a. If no tasks have been created yet, the dashboard shows an empty state with the prompt: 'No tasks yet. Create your first task to get started!'

Task 3: Requirements Elicitation and Analysis

Elicitation Technique

The primary elicitation techniques used for TaskFlow Pro were Brainstorming and Analysis of Existing Systems.

Brainstorming was used as the initial technique because the project concept spans multiple domains — software development, marketing, HR, education, design, and more. Structured brainstorming sessions allowed the team to enumerate all desirable features from the perspectives of different types of managers and team members across these domains. This technique was particularly appropriate because it is fast, domain-agnostic, and surfaces a wide variety of use cases without requiring access to a large external user base.

Analysis of Existing Systems was conducted by studying well-established tools: Google Classroom (for the assignment and resource-sharing model), LeetCode (for the coding challenge execution concept), Jira and Trello (for the task management dashboard

model), and Microsoft Teams Assignments (for cross-domain task workflows). By analyzing these tools, the team identified a clear gap: no existing platform combines universal, domain-agnostic task assignment with an integrated code compiler and AI-powered deliverable feedback in a team-scoped workspace. This analysis directly informed the task type tag system, the flexible submission format model, and the AI feedback layer that adapts its analysis based on task type.

Stakeholder Identification

Primary Stakeholders and Pain Points:

- Team Managers and Project Leaders (any domain): Pain point — fragmented tooling. A marketing manager uses one tool for assigning briefs, another for tracking deadlines, and another for sharing resources. A software engineering lead uses a project tracker, a separate code review tool, and email for materials. TaskFlow Pro consolidates all of these into a single workspace, reducing context-switching and tool overhead.
- Software / Engineering Team Members: Pain point — no integrated platform where they can receive a coding task, write and run code, and get automated feedback all in one place. TaskFlow Pro's built-in compiler and AI code review directly address this.
- Non-Technical Team Members (HR, Marketing, Finance, Design): Pain point — most task management platforms are designed with software teams in mind and feel over-engineered for non-technical workers. TaskFlow Pro's flexible task types, plain-text and file-based submissions, and intuitive interface make it equally accessible to non-technical users.
- Educators and Student Teams: Pain point — lack of a team-scoped assignment platform that handles both coding assignments and general academic tasks in one place. TaskFlow Pro serves as a lightweight, AI-enhanced alternative to Google Classroom with the added benefit of an integrated compiler for programming courses.

Task 4: Gradio/Streamlit Demo

Demo Overview

The Gradio demo showcases the core ML functionality of TaskFlow Pro: AI-powered task feedback. The demo accepts a task type (Coding, Document, Marketing, HR, etc.) and a submitted deliverable (code or text), and returns an AI-generated quality score, identified issues, and improvement suggestions — demonstrating how the system provides intelligent feedback across any task domain.

Demo URLs

Public Gradio Demo URL:

https://d17bbe34215cdd61ef.gradio.live

Google Colab Notebook URL:

Google_colab_TaskFlow

Handwritten Document

[Scan of handwritten document containing Project Introduction (Purpose) and One Use Case to be inserted here as an image]

Assignment 2

Project : Task Flow Pro

Purpose :-

The purpose of Task Flow is to provide a centralized, AI enhanced workspace where team leaders can assign, track, and manage any kind of task of their team members — regardless of the department or nature of the work. Unlike tools that are narrow in focus, Task Flow Pro is built as a universal task assigner inspired by the flexibility of Google Classroom. A manager can assign a software engineer to fix a bug, ask a designer to create a graph or a infographic, instruct an HR associate to draft a policy document, requesting a marketing analyst to prepare a campaign report, or set a finance intern to complete a spreadsheet analysis — all from the same interface.

Use Case:
Assign a Task to team Members.

Use Case ID: UC-03
Use Case Name: Assign a Task to team members
Actors: Manager
Preconditions: The manager is logged in, belongs to a Team, and at least one team member exists in the team.
Post conditions: A new task is created, stored, and associated with the selected team members, In app, notification are sent to assigned members

Main Flow
- The manager navigates to the Create Task Page.
- The manager fills in Task title and a detailed description.
- The manager selects a Task type tag from the available options: Coding, Design, Document, Research, Marketing or Custom.
- If test type is Coding, the manager additionally selects the target language and optionally enter test cases.
- For all other task, manager provide any relevant instruction or expected output format in the description.
- The manager sets a deadline
- The manager can optionally attach

attach reference materials (PDFs, URLs, Docs).
- The manager clicks assign Task.
- The system validates all required fields and confirm deadline is in the future
- The system saves the task records and sends in-app notification to all assigned members.

Alternative Flows:
- If the deadline is in the past, the system displays: 'Deadline must be a future Date and time'.

- If no team members are selected, the system displays: 'Please select at least one team member'.